

Recommendations for Life History Data Needs for US Caribbean Fisheries Stock Assessment

Summary

A primary goal of the Southeast Fisheries Science Center's Caribbean Strategic Planning project is to reduce gaps in life history datasets used to inform management. The first step in accomplishing that goal is to identify major life history data gaps and information needed to improve fisheries management decisions in the U.S. Caribbean. A working group was established to identify priority species for the U.S. Caribbean, identify available life history data including historical and ongoing efforts, summarize the identified life history data sources, identify life history data needs by species and spatial extent, and provide recommendations to the stakeholders to implement data collection that fill the gaps and inform stock assessment using the most currently available data. This document contains those findings.



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Acronyms

<i>Acronym</i>	<i>Definition</i>
CIMAS	Cooperative Institute for Marine and Atmospheric Studies
CRP	Cooperative Research Program grant
FMP	Fishery Management Plan
MARFIN	Marine Fisheries Initiative grant
PR	Puerto Rico
PR DRNA	Puerto Rico Departamento de Recursos Naturales y Ambientales
SEAMAP-C	Southeast Area Monitoring and Assessment Program - Caribbean
SEDAR	Southeast Data Assessment and Review
SEFSC	Southeast Fisheries Science Center
SK	Saltonstall-Kennedy grant
STJ	St. John
STT	St. Thomas
USC-A	University of South Carolina-Aiken
USVI	United States Virgin Islands

Working Group

<i>Participant</i>	<i>Email</i>	<i>Role</i>	<i>Affiliation</i>
Robert Allman*	–	Member	SEFSC
Rachel Banton	rachel.banton@noaa.gov	Coordinator	CIMAS
Stephanie Martinez	stephanie.martinez@noaa.gov	Member	SEFSC
Noemi Peña	nopena@drna.pr.gov	Lead	PR DRNA
Jesús Rivera Hernández	jriverahernandez.pr@gmail.com	Member	USC-A
Michelle Sharer-Umpierre	ciel.uprm@gmail.com	Member	HJR Reefscaping
Virginia Shervette	shervette@gmail.com	Member	USC-A

*Participated in the working group until his departure from federal service.

Please contact a working group member with changes or suggestions.

Introduction

The data essential to complete a fisheries stock assessment include the abundance, biology, and catch for the species. Biological data refers to life history demographics including parameters for weight-length relationships, age-length relationships, maximum age, growth, maturity, and mortality. These parameters are inputted into the selected assessment model to inform the health of the population given the level of catch observed in the fishery. It is preferred that the life history parameters are obtained through research conducted on samples from the region in which the stock is being assessed, rather than from other regions (i.e., for a Puerto Rico stock assessment, it is best if life history parameters come from Puerto Rico rather than Florida). If data from the region are not available at the time of the assessment, estimates are made using data from nearby regions. This increases the uncertainty in the assessment model and the results may not be as informative for management. This working group was formed to strengthen the stock assessment enterprise in the U.S. Caribbean and improve the data available for stock assessment by identifying species-specific life history data gaps in the region.

Participants volunteered to join this strategic planning working group. During the initial meeting of the group, the participants developed their purpose, goals, and ideal outcomes, which are documented in the [Team Charter](#). The group met sixteen times over the course of two years to accomplish the goals. Section A contains those goals and what was accomplished under each goal. Section B builds on the accomplished goals and contains specific recommendations for life history data collection.

A. Goals & Summary of Actions to Address Goals:

1. Identify priority species for the U.S. Caribbean

Many species are included in the island-based FMPs, but in tackling this project it was important to identify which species were considered priority in terms of identifying available life history data. The SEFSC Caribbean Fisheries Branch Data Triage Project categorized species as having a likelihood for a successful stock assessment based on the availability of the data required for the assessment (e.g., abundance, biology, and catch). Results of the data triage project were referenced to identify priority species lists for each island platform (i.e. Puerto Rico, St. Thomas/St. John, and St. Croix). The island-based FMPs were also referenced and species assigned as “indicator” were automatically included on the lists. Species in which ongoing life history research is underway were also included on the priority lists.

2. Identify available life history data for priority species

Group discussion and literature review indicated that existing inventories of Caribbean life history data sources were non-comprehensive. The SEFSC Caribbean Fisheries Branch Data Triage Project and Stevens et al., 2019 have addressed cataloging reliable life history data sources and parameters in the past. Other forums have also worked to identify life history data sources including the SEFSC Caribbean Research Inventory and Fish/Fisheries Conservation Lab at USC-A. These efforts have identified a number of reliable life history sources available at the time, but a comprehensive catalog was still needed.

A new and comprehensive inventory was compiled to catalog reliable life history data for U.S. Caribbean priority species by referencing the previous efforts, performing additional literature searches, and soliciting feedback from life history researchers in the region. The inventory sought to capture the publication link, species, sampling location and years, depth range of sampling, gear used, sample size, and type of life history data collected and analyzed such as length age, length length, length weight, reproduction/maturity, and mortality. See Table 1 for the list of life history publications and the [associated spreadsheet](#) for additional information.

3. Summarize available data and identify life history data needs by species and spatial extent

The data gap identification process involved many steps. First, separate tables were created for each island platform and included each of the associated priority species. Next, it was noted which species had an assessment attempted in the SEDAR process. Throughout SEDAR, the lead analyst will compile a list of recommendations as to what data, if they were available, would improve the assessment modeling and better inform the management advice. These recommendations are listed at the end of the assessment report and were included in the tables. Next, the SEDAR recommendations and inventory of published data were used to summarize the current status of available life history data and the data collected and made available since the previous assessment. For species in which a SEDAR has not been attempted, the inventory was used to inform the status of available data. See Tables 2–4 for the information on SEDAR recommendations, overall status of available data, and availability of reproduction and age/growth data. Finally, a table highlighting the existing data gaps was produced. See Table 5.

4. Provide recommendations to the stakeholders to implement data collections that fill the data gaps and inform stock assessment using the most current available data

This working group concisely documented what data collections are needed to inform stock assessment and management advice. This present report, specifically section C, contains the findings of this group.

Table 1. Life history publications of priority species in the U.S. Caribbean.

<i>SPECIES</i>	<i>PLATFORM</i>	<i>YEARS</i>	<i>DATA TYPE</i>	<i>PUBLICATION</i>
Queen triggerfish	PR, USVI	2013-2023	Length age, reproduction/maturity	Rivera-Hernández & Shervette, 2024
Stoplight parrotfish, queen parrotfish	PR, USVI	2018-2019	Length age	Rivera-Hernández & Shervette, 2024
Stoplight parrotfish, queen parrotfish	PR, USVI	2015-2023	Length age, reproduction/maturity	Rivera-Hernández & Shervette, 2024
Yellowtail snapper	PR, USVI	2009-2023	Length age, length length, length weight, reproduction/maturity	Shervette et al., 2024
Yellowtail snapper	PR, USVI	2010-2023	Length age, length weight	Shervette & Rivera Hernandez, 2024
Red hind	USVI	2018-2019	Reproduction/maturity	Rosemond et al., 2022
Queen triggerfish	PR, USVI	2013-2021	Length age	Shervette & Rivera Hernandez, 2022
Caribbean spiny lobster	Wider Caribbean	2017-2019	Reproduction/maturity	Atherley et al., 2021
Princess parrotfish	PR, USVI	2015-2020	Length age, reproduction/maturity	Jones et al., 2021
Yellowfin grouper	STT	2008, 2009, 2011, 2014	Reproduction/maturity	Nemeth et al., 2020
Queen triggerfish	PR, STX	2013-2018	Reproduction/maturity	Rivera-Hernández et al., 2018
Caribbean spiny lobster	STT	2012-2016	Length age	Olsen et al., 2017
Caribbean spiny lobster	Wider Caribbean	2015-2016	Reproduction/maturity	Tewfik & Babcock, 2017
Blackfin snapper	PR, USVI	1979-2015	Length weight	Burton et al., 2016
Hogfish	Wider Caribbean	2011-2013	Length length, length weight	Pérez-Chacón, 2015
Yellowtail snapper	USVI	1974-2012	Length age, mortality	Olsen, 2014
French grunt, schoolmaster snapper	PR, STX	2007	Length age	Mateo et al., 2011
Stoplight parrotfish	Wider Caribbean		Length length	Molloy et al. 2011
Yellowfin grouper	STT	2004-2009	Reproduction/maturity	Kadison et al., 2010
Wahoo	Wider Caribbean	1997-2006	Reproduction/maturity	Maki Jenkins & McBride, 2009
Snappers, groupers, hogfish	PR	1985-1987, 2000-2005	Length age, reproduction/maturity, mortality	Ault et al., 2008
King mackerel, cero mackerel	PR	2001-2002	Reproduction/maturity	Figueroa-Fernández et al., 2007
Caribbean spiny lobster, queen conch	Wider Caribbean			Posada & Glazer, 2007

Silk snapper	PR	2005	Reproduction/maturity	Rosario et al., 2006
Reef fish	STT	2005-2006	Length weight	Trumble et al., 2006
Caribbean spiny lobster	PR	1980-2003		Chormanski et al., 2005
Queen snapper	Wider Caribbean	1986-1987, 1992, 1999-2000, 2001	Length length, length weight, Reproduction/maturity	Gobert et al., 2005
Red hind	USVI	1999-2004	Reproduction/maturity	Nemeth, 2005
Dolphinfish	PR	1990-1992	Reproduction/maturity	Perez & Roman, 2005
Red hind	STT	2002-2003	Reproduction/maturity	Whiteman et al., 2005
Stoplight parrotfish	Wider Caribbean	1995-2000	Length age	Choat et al. 2003
Wahoo	Wider Caribbean	1976-2000	Length age, length weight, Reproduction/maturity	Oxenford et al., 2003
Parrotfish, surgeonfish	PR, USVI	1991-2000	Length weight	Pagan, 2002
Coney grouper	PR	1997-1999	Reproduction/maturity	Figueroa et al., 2001
Mutton snapper	PR	2000-2002	Reproduction/maturity	Figueroa & Torres., 2001
Wahoo	Wider Caribbean	1997-1998	Reproduction/maturity	Brown-Peterson et al., 2000
Wahoo	Wider Caribbean	1997-1998	Length age	Franks et al., 2000
Lane snapper	Wider Caribbean	1992-1996	Length age	Luckhurst et al., 2000
Dolphinfish	PR	1991-1992	Length age, length weight	Rivera & Appeldoorn, 2000
Caribbean spiny lobster	STX	1995-1999	Length age	Mateo & Tobias, 1999
Groupers, parrotfish, snappers	PR	1996-1998	Reproduction/maturity	Figueroa et al., 1998
Stoplight parrotfish	Wider Caribbean	1989-1992	Mortality	van Rooij & Videler, 1997
Mutton snapper	Wider Caribbean	1991-1992	Length length	Mueller, 1995
Coney grouper, red hind	USVI	1978-1992	Length weight	Bolden, 1994
Dolphinfish	PR		Length age, length weight	Rivera Betancourt, 1994
Red hind	PR	1987-1992	Reproduction/maturity	Sadovy et al., 1994
Red hind	PR	1982-1989	Length weight, reproduction/maturity	Shapiro et al., 1993
Lane snapper	PR	1988-1988	Length age, length weight, mortality	Acosta & Appeldoorn., 1992
Red hind	USVI	1984-1991	Length weight	Beets & Friedlander, 1992
Red hind	Wider Caribbean	1990	Length weight	Luckhurst, 1992
Dolphinfish	PR	1990-1992	Length weight, reproduction/maturity	Pérez et al., 1992

Red hind	PR, STT	1987-1989	Length weight, reproduction/maturity	Sadovy et al., 1992
Silk snapper	PR	1989-1991	Reproduction/maturity	Figueroa, 1991
Dolphinfish	PR	1990-1991	Length weight, reproduction/maturity	Perez & Sadovy, 1991
White grunt	PR	1988-1989		Román, 1991
Red hind, white grunt	PR, STT		Length age	Sadovy et al., 1989
Reef fish	PR, USVI	1985	Length weight	Bohnsack & Harper, 1988
Lane snapper	Wider Caribbean	1979-1981	Length age, reproduction/maturity	Manickchand-Dass, 1987
Yellowtail snapper, queen triggerfish	PR, USVI	1983-1984	Length age, length weight	Manooch & Drennon, 1987
Yellowfin grouper	STJ		Length age	Munro & Williams, 1985
Hogfish	PR		Reproduction/maturity	Colin, 1982
Snappers	PR	1976-79	Length weight, reproduction/maturity	Boardman & Weiler, 1980
White grunt	PR	1973	Length age	Stevenson, 1978
Silk snapper	USVI	1970-1972	Length weight	Sylvester & Dammann, 1973
Groupers	STJ	1959-1960	Length age	Randall, 1962

Table 2. Comprehensive list of available life history data for priority species in Puerto Rico.

SPECIES	SEDAR	SEDAR DATA RECOMMENDATION	STATUS OF AVAILABLE DATA	AVAILABLE DATA	
				REPRODUCTION	AGE / GROWTH
Bar jack	–	–	Non-comprehensive	Scarce data from SEAMAP-C	–
Blackfin tuna	–	–	Non-comprehensive	–	–
Blue runner	–	–	Non-comprehensive	Final report from F48 (2003-07) and data from SEAMAP-C	–
Boxfishes	–	–	Non-comprehensive	Scarce data from SEAMAP-C (only <i>L. trigonus</i>) /Final Report Yvone Sadovy, 1990	Scarce data from SEAMAP-C (only <i>L. trigonus</i>) /Final Report Yvone Sadovy, 1990
Caribbean spiny lobster	SEDAR8 SEDAR57 SEDAR91	Mortality of juveniles Size composition Stock structure and abundance Size composition Life history [IN PROGRESS]	Non-comprehensive	–	–
Cero mackerel	–	–	Non-comprehensive	Figueroa-Fernández et al., 2007	–
Coney grouper	–	–	Non-comprehensive	Some reproductive histology data from SEAMAP-C samples	–
Dolphinfish	–	–	Non-comprehensive	DRNA opportunistic tournament sampling/ Perez & Sadovy, 1991	Rivera & Appeldoorn, 2000
Graysby grouper	–	–	Non-comprehensive	Some reproductive histology data from SEAMAP-C samples	–
Hogfish	SEDAR46	Length composition Life history	Comprehensive	CRP report 2015-2021 collections for life history	CRP report 2015-2021 collections for life history
King mackerel	–	–	Non-comprehensive	Figueroa-Fernández et al., 2007	–
Lane snapper	–	–	Comprehensive	2012-2023 collections for life history; Figueroa et al., 1998	2012-2023 collections for life history
Little tunny	–	–	Non-comprehensive	–	–

Mutton snapper	SEDAR14	Size composition Age composition Life history	Non-comprehensive	Figueroa & Torres., 2001 ; Data from 2016-2022	–
Queen conch	SEDAR14	Size composition Age composition Habitat	Non-comprehensive	SK 2024-2026 collections for life history	SK 2024-2026 collections for life history
Queen parrotfish	–	–	Comprehensive	Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024	Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024
Queen snapper	SEDAR4 SEDAR26	[NONE] Age and growth Life history Size composition	Non-comprehensive	SK 2023-2025 life history study underway	SK 2023-2025 life history study underway
Queen triggerfish	SEDAR30 SEDAR80	Size composition Age composition Life history Growth Life history	Comprehensive	Rivera-Hernandez & Shervette, 2024 covers 2013-2023	Rivera-Hernandez & Shervette, 2024 covers 2013-2023
Rainbow parrotfish	–	–	Non-comprehensive	–	–
Rainbow runner	–	–	Non-comprehensive	–	–
Red hind grouper	SEDAR35	Size composition Age composition Life history	Non-comprehensive	Some reproductive histology data from SEAMAP-C samples	–
Redtail parrotfish	SEDAR26	Size composition Age composition Age and growth Life history	Comprehensive	Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples; Figueroa et al., 1998	Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples
Silk snapper	SEDAR26	Size composition Age composition Age and growth Life history	Non-comprehensive	SK 2023-2025 life history study underway	SK 2023-2025 life history study underway
Skipjack tuna	–	–	Non-comprehensive	–	–
Stoplight parrotfish	SEDAR84	[IN PROGRESS]	Comprehensive	Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024 ; Figueroa et al., 1998	Life history data available from 2013-2023, SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024

Wahoo	–	–	Non-comprehensive	DNER opportunistic tournament sampling/Final Report F48 (2004-2009)	–
White grunt	–	–	Non-comprehensive	Some reproductive histology data from SEAMAP-C samples; Román, 1991	Sadovy et al., 1989
Yellowfin tuna	–	–	Non-comprehensive	–	–
Yellowtail snapper	SEDAR8 SEDAR46 SEDAR84	Movement patterns Growth Life history Mortality Size composition Age composition Length composition Life history [IN PROGRESS]	Comprehensive	SEDAR 84 Report (working paper); Figueroa et al., 1998	SEDAR 84 Report (working paper)

Table 3. Comprehensive list of available life history data for priority species in St. Thomas / St. John.

SPECIES	SEDAR	SEDAR DATA RECOMMENDATION	STATUS OF AVAILABLE DATA	AVAILABLE DATA	
				REPRODUCTION	AGE / GROWTH
Blackfin snapper	–	–	Non-comprehensive	–	–
Blue runner	–	–	Non-comprehensive	–	–
Blue tang	SEDAR30	Size composition Age composition Life history	Non-comprehensive	–	–
Boxfishes	–	–	Non-comprehensive	–	–
Caribbean spiny lobster	SEDAR8	Mortality of juveniles Size composition	Non-comprehensive	–	–
	SEDAR46	Length composition Life history			
	SEDAR57	Stock structure and abundance Size composition Life history			
	SEDAR91	[IN PROGRESS]			
Cero mackerel	–	–	Non-comprehensive	–	–
Coney grouper	–	–	Non-comprehensive	–	–
Doctorfish	–	–	Non-comprehensive	–	–
Dolphinfish	–	–	Non-comprehensive	–	–
Gray angelfish	–	–	Non-comprehensive	–	–
Graysby grouper	–	–	Non-comprehensive	–	–
Hogfish	–	–	Non-comprehensive	–	–
King mackerel	–	–	Non-comprehensive	–	–
Lane snapper	–	–	Comprehensive	2012-2023 collections for life history	2012-2023 collections for life history
Little tunny	–	–	Non-comprehensive	–	–
Margate grunt	–	–	Non-comprehensive	–	–

Mutton snapper	SEDAR14	Size composition Age composition Life history	Non-comprehensive	–	–
Queen angelfish	–	–	Non-comprehensive	–	–
Queen conch	SEDAR14	Size composition Age composition Habitat	Non-comprehensive	–	–
Queen snapper	SEDAR26	Age and growth Life history Size composition	Non-comprehensive	SK 2023-2025 life history study underway	SK 2023-2025 life history study underway
Queen triggerfish	SEDAR30	Size composition Age composition Life history	Comprehensive	Rivera-Hernández & Shervette, 2024 covers 2013-2023	Rivera-Hernández & Shervette, 2024 covers 2013-2023
	SEDAR46	Length composition Life history			
	SEDAR80	[IN PROGRESS]			
Rainbow runner	–	–	Non-comprehensive	–	–
Red hind grouper	SEDAR35	Size composition Age composition Life history	Non-comprehensive	CRP 2024-2025 study initiated	CRP 2024-2025 study initiated
Redtail parrotfish	SEDAR26	Size composition Age composition Age and growth Life history	Comprehensive	Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples	Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples
Scrawled cowfish	–	–	Non-comprehensive	–	–
Stoplight parrotfish	–	–	Comprehensive	Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024 ; Rivera-Hernández & Shervette, 2024	Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024 ; Rivera-Hernández & Shervette, 2024
Wahoo	–	–	Non-comprehensive	–	–
White grunt	–	–	Non-comprehensive	–	–

Table 4. Comprehensive list of available life history data for priority species in St. Croix.

SPECIES	SEDAR	SEDAR DATA RECOMMENDATION	STATUS OF AVAILABLE DATA	AVAILABLE DATA	
				REPRODUCTION	AGE / GROWTH
Blackfin snapper	–	–	Non-comprehensive	–	–
Blue tang	SEDAR30	Size composition Age composition Life history	Non-comprehensive	–	–
Bluestriped grunt	–	–	Non-comprehensive	–	–
Caribbean spiny lobster	SEDAR8	Mortality of juveniles Size composition	Non-comprehensive	–	–
	SEDAR46	Length composition Life history			
	SEDAR57	Stock structure and abundance Size composition Life history			
	SEDAR91	[IN PROGRESS]			
Coney grouper	–	–	Non-comprehensive	–	–
Doctorfish	–	–	Non-comprehensive	–	–
Dolphinfish	–	–	Non-comprehensive	–	–
French angelfish	–	–	Non-comprehensive	–	–
Gray angelfish	–	–	Non-comprehensive	–	–
Honeycomb cowfish	–	–	Non-comprehensive	–	–
King mackerel	–	–	Non-comprehensive	–	–
Little tunny	–	–	Non-comprehensive	–	–
Mutton snapper	SEDAR14	Size composition Age composition Life history	Non-comprehensive	–	–
Princess parrotfish	–	–	Non-comprehensive	Jones et al., 2021	Jones et al., 2021
Queen conch	SEDAR14	Size composition Age composition Habitat	Non-comprehensive	SK 2024-2026 collections for life history	SK 2024-2026 collections for life history

Queen parrotfish	–	–	Comprehensive	Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024 ; Rivera-Hernández & Shervette, 2024	Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024 ; Rivera-Hernández & Shervette, 2024
Queen triggerfish	SEDAR30 SEDAR80	Size composition Age composition Life history [IN PROGRESS]	Comprehensive	Rivera-Hernández & Shervette, 2024 covers 2013-2023	Rivera-Hernández & Shervette, 2024 covers 2013-2023
Red hind grouper	SEDAR35	Size composition Age composition Life history	Non-comprehensive	–	–
Redband parrotfish	–	–	Comprehensive	Life history data available from 2013-2023; MARFIN 2021 report	Life history data available from 2013-2023; MARFIN 2021 report
Redfin parrotfish	–	–	Comprehensive	Life history data available from 2013-2023; MARFIN 2021 report	Life history data available from 2013-2023; MARFIN 2021 report
Redtail parrotfish	SEDAR26	Size composition Age composition Age and growth Life history	Comprehensive	Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples	Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples
Scrawled cowfish	–	–	Non-comprehensive	–	–
Stoplight parrotfish	SEDAR46 SEADR84	Length composition Life history [IN PROGRESS]	Comprehensive	Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernandez & Shervette, 2024 ; Rivera-Hernandez & Shervette, 2024	Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernandez & Shervette, 2024 ; Rivera-Hernandez & Shervette, 2024
White grunt	–	–	Non-comprehensive	–	–
Yellowfin tuna	–	–	Non-comprehensive	–	–
Yellowtail snapper	SEDAR8	Growth Life history Mortality Size composition Age composition	Non-comprehensive	–	–

Table 5. Life history data needs for priority species in the U.S. Caribbean. X indicates no data are available. O indicates some data are available but more are still needed. — indicates the species is not a priority for that island platform. Blanks indicate data are available. Species for which data availability was listed as "comprehensive" in Tables 2–4 were removed from this table.

SPECIES	PUERTO RICO DATA NEEDS		ST. THOMAS / ST. JOHN DATA NEEDS		ST. CROIX DATA NEEDS	
	REPRODUCTION	AGE / GROWTH	REPRODUCTION	AGE / GROWTH	REPRODUCTION	AGE / GROWTH
Bar jack	O	X	—	—	—	—
Blackfin snapper	—	—	X	X	X	X
Blackfin tuna	X	X	—	—	—	—
Blue runner	O	X	X	X	—	—
Blue tang	—	—	X	X	X	X
Bluestriped grunt	—	—	—	—	X	X
Boxfishes	O	O	X	X	—	—
Caribbean spiny lobster	X	X	X	X	X	X
Cero mackerel	O	X	X	X	—	—
Coney grouper	O	X	X	X	X	X
Doctorfish	—	—	X	X	X	X
Dolphinfish	O	X	X	X	X	X
French angelfish	—	—	—	—	X	X
Gray angelfish	—	—	X	X	X	X
Graysby grouper	O	X	X	X	—	—
Hogfish			X	X	—	—
Honeycomb cowfish	—	—	—	—	X	X
King mackerel	O	X	X	X	X	X
Lane snapper					—	—
Little tunny	X	X	X	X	X	X
Mutton snapper	O	X	X	X	X	X
Princess parrotfish	—	—	—	—	X	X
Queen angelfish	—	—	X	X	—	—
Queen conch	O	O	O	O	O	O
Queen parrotfish			—	—		
Queen snapper	O	O	O	O	—	—

Queen triggerfish						
Rainbow parrotfish	X	X	—	—	—	—
Rainbow runner	X	X	X	X	—	—
Red hind grouper	O	X	O	O	X	X
Redband parrotfish	—	—	—	—		
Redfin parrotfish	—	—	—	—		
Redtail parrotfish						
Scrawled cowfish	—	—	X	X	X	X
Silk snapper	O	O	—	—	—	—
Skipjack tuna	X	X	—	—	—	—
Stoplight parrotfish						
Wahoo	O	X	X	X	—	—
White grunt	O	O	X	X	X	X
Yellowfin tuna	X	X	—	—	X	X
Yellowtail snapper			—	—	X	X

B. Availability of Life History Data:

1. Puerto Rico Life History Data Availability

Among the priority species list for Puerto Rico, seven species (hogfish, lane snapper, queen parrotfish, queen triggerfish, redband parrotfish, stoplight parrotfish, and yellowtail snapper) have comprehensive regionally-specific life history data available for use in stock assessment, 15 species have some or temporally limited data available, and seven do not have comprehensive data available. The following priority species have at least some reproductive data available: bar jack, blue runner, cero mackerel, coney grouper, graysby grouper, king mackerel, mutton snapper, red hind grouper, and wahoo. The following priority species/groups have at least some reproductive and age/growth data available: boxfishes, dolphinfish, hogfish, lane snapper, queen conch, queen parrotfish, queen snapper, queen triggerfish, redband parrotfish, silk snapper, stoplight parrotfish, white grunt, and yellowtail snapper. The following priority species do not have life history data available: blackfin tuna, Caribbean spiny lobster, little tunny, rainbow parrotfish, rainbow runner, skipjack tuna, and yellowfin tuna.

2. St. Thomas/St. John Life History Data Availability

Among the priority species list for St. Thomas/St. John, four species (lane snapper, queen triggerfish, redband parrotfish, and stoplight parrotfish) have comprehensive regionally-specific life history data available for use in stock assessment, three species have some or temporally limited data available, and 24 do not have comprehensive data available. The following priority species have at least some reproductive and age/growth data available: lane snapper, queen conch, queen snapper, queen triggerfish, red hind grouper, redband parrotfish, stoplight parrotfish. The following priority species/groups do not have life history data available: blackfin snapper, blue runner, blue tang, boxfishes, Caribbean spiny lobster, cero mackerel, coney grouper, doctorfish, dolphinfish, gray angelfish, graysby grouper, hogfish, king mackerel, little tunny, margate grunt, mutton snapper, queen angelfish, rainbow runner, scrawled cowfish, wahoo, and white grunt.

3. St. Croix Life History Data Availability

Among the priority species list for St. Croix, six species (queen parrotfish, queen triggerfish, redband parrotfish, redband parrotfish, redband parrotfish, and stoplight parrotfish) have comprehensive regionally-specific life history data available for use in stock assessment, one species has some or temporally limited data available, and 19 do not have comprehensive data available. The following priority species have at least some reproductive and age/growth data available: queen conch, queen parrotfish, queen triggerfish, redband grouper, redband parrotfish, redband parrotfish, and stoplight parrotfish. The following priority species/groups do not have life history data available: blackfin tuna, blue tang, bluestriped grunt, Caribbean spiny lobster, coney grouper, doctorfish, dolphinfish, french angelfish, gray angelfish, honeycomb cowfish, king mackerel, little tunny, mutton snapper, princess parrotfish, red hind grouper, scrawled cowfish, white grunt, yellowfin tuna, and yellowtail snapper.

C. Specific Recommendations for Life History Data Needs:

1. Puerto Rico Life History Data Needs

Life history data from Puerto Rico for priority species are not currently comprehensive. In order to obtain informative management advice from stock assessment methods, additional biological data are essential. It is important that reproductive and age and growth data are collected island wide from Puerto Rico for the following priority species: bar jack, blackfin tuna, blue runner, boxfishes, Caribbean spiny lobster, cero mackerel, coney grouper, dolphinfish, graysby grouper, king mackerel, little tunny, mutton snapper, queen conch, queen snapper, rainbow parrotfish, rainbow runner, red hind grouper, silk snapper, skipjack tuna, wahoo, white grunt, and yellowfin tuna. For the species in which currently available life history data may be sufficient for a successful stock assessment, continued reproductive and age and growth data collections should still continue to inform future management actions. Those priority species include hogfish, lane snapper, queen parrotfish, queen triggerfish, redband parrotfish, stoplight parrotfish, and yellowtail snapper.

2. St. Thomas / St. John Life History Data Needs

Life history data from St. Thomas/St. John for priority species are not currently comprehensive. In order to obtain informative management advice from stock assessment methods, additional biological data are essential. It is important that reproductive and age and growth data are collected island wide from St. Thomas/St. John for the following priority species: blackfin snapper, blue runner, blue tang, boxfishes, Caribbean spiny lobster, cero mackerel, coney grouper, doctorfish, dolphinfish, gray angelfish, graysby grouper, hogfish, king mackerel, little tunny, margate grunt, mutton snapper, queen angelfish, queen conch, queen snapper, rainbow runner, red hind grouper, scrawled cowfish, wahoo, and white grunt. For the species in which currently available life history data may be sufficient for a successful stock assessment, continued reproductive and age and growth data collections should still continue to inform future management actions. Those priority species include lane snapper, queen triggerfish, redband parrotfish, and stoplight parrotfish.

3. St. Croix Life History Data Needs

Life history data from St. Croix for priority species are not currently comprehensive. In order to obtain informative management advice from stock assessment methods, additional biological data are essential. It is important that reproductive and age and growth data are collected island wide from St. Croix for the following priority species: blackfin snapper, blue tang, bluestriped grunt, Caribbean spiny lobster, coney grouper, doctorfish, dolphinfish, french angelfish, gray angelfish, honeycomb cowfish, king mackerel, little tunny, mutton snapper, princess parrotfish, queen conch, red hind grouper, scrawled cowfish, white grunt, yellowfin tuna, and yellowtail snapper. For the species in which currently available life history data may be sufficient for a successful stock assessment, continued reproductive and age and growth data collections should still continue to inform future management actions. Those priority species include queen parrotfish, queen triggerfish, redband parrotfish, redfin parrotfish, redband parrotfish, and stoplight parrotfish.

Appendix

- Acosta, A. & R.S. Appeldoorn. 1992. Estimation of Growth Mortality and Yield Per Recruit For *Lutjanus Synagris* (Linnaeus) in Puerto Rico. *Bulletin of Marine Science*, 50(2): 282–291.
- Atherley, N.A.M., M.M. Dennis, D.C. Behringer, M.A. Freeman. 2021. Size at sexual maturity and seasonal reproductive activity of the Caribbean spiny lobster *Panulirus argus*. *Mar Ecol Prog Ser*, 671: 129–145.
- Ault, J.S., S.G. Smith, J. Luo, M.E. Monaco, R.S. Appeldoorn. 2008. Length-based assessment of sustainability benchmarks for coral reef fishes in Puerto Rico. *Environmental Conservation*, 35(3): 221–231.
- Beets, J. & A. Friedlander. 1992. Stock analysis and management strategies for red hind, *Epinephelus guttatus*, in the U.S. Virgin Islands. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 42: 66–79.
- Boardman, C. & D. Weiler. 1980. Aspects of the life history of three deepwater snappers around Puerto Rico. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 32: 158–172.
- Bohnsack, J.A. & D.E. Harper. 1988. Length-weight relationships of selected marine reef fishes from the southeastern United States and the Caribbean. NOAA Technical Memorandum NMFS-SEFC-215, 31 p.
- Bolden, S.K. 1994. A summary of biological and fishery data on red hind (*Epinephelus guttatus*) and coney (*Cephalopholis fulva*) stocks in the U.S. Virgin Islands. Contribution: MIA-93/94-32, NOAA/NMFS/SEFSC, 33 p.
- Brown-Peterson, N.J., J.S. Franks, A.M. Burke. 2000. Preliminary observations on the reproductive biology of Wahoo, *Acanthocybium solandri*, from the northern Gulf of Mexico and Bimini, Bahamas. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 51: 414–427.
- Burton, M.L., J.C. Potts, D.R. Carr. 2016. Age, Growth and Natural Mortality of Blackfin Snapper *Lutjanus buccanella* from the Southeastern United States and U. S. Caribbean. *Gulf and Caribbean Research* 27 (1): 66–73.
- Caribbean Fishery Management Council, 2019. Comprehensive Fishery Management Plan for the Puerto Rico Exclusive Economic Zone. 637 p.
[https://caribbeanfmc.com/FMP_Island_Based_2019/EA_FMP_Puerto_Rico_Final.pdf]
- Caribbean Fishery Management Council, 2019. Comprehensive Fishery Management Plan for the St. Croix Exclusive Economic Zone. 509 p.
[https://caribbeanfmc.com/FMP_Island_Based_2019/EA_FMP_St_Croix_Final.pdf]
- Caribbean Fishery Management Council, 2019. Comprehensive Fishery Management Plan for the St. Thomas/St. John Exclusive Economic Zone. 507 p.
[https://caribbeanfmc.com/FMP_Island_Based_2019/EA_FMP_St_Thomas.St_John_Final.pdf]
- Choat, J.H., D.R. Robertson, J.L. Ackerman, J.M. Posada. 2003. An age-based demographic analysis of the Caribbean stoplight parrotfish *Sparisoma viride*. *Mar Ecol Prog Ser*, 246: 265–277.
- Chormanski, S.D., D. Die, S. Saul. 2005. Length Frequency Analysis of Caribbean Spiny Lobster (*Panulirus argus*) Sampled by the Puerto Rico Commercial Trip Interview Program (1980 – 2003). SEDAR8-RW-02, 27 p.
- Colin, P.L. 1982. Spawning and larval development of the hogfish, *Lachnolaimus maximus* (Pisces: Labridae). *Fishery Bulletin*, 80(4): 853–862.
- Figueroa, M. & W. Torres. 2001. Aspectos de la biología reproductiva de la sama (*Lutjanus analis*) en Puerto Rico y recomendaciones para su manejo. Informe Final del Proyecto Titulado. 30 p.
- Figueroa, M., W. Torres, A. Rosario. 2001. Reproducción en el mero mantequilla (*Cephalopholis fulva*) en Puerto Rico. 32 p.
- Figueroa, M., W. Torres, N. Peña. 2007. Sexual maturity and reproductive seasonality of king mackerel (*S. cavalla*) and cero (*S. regalis*) in Puerto Rico. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 58: 251–261.
- Figueroa, M., D. Matos-Caraballo, W. Torres. 1998. Maturation and reproductive seasonality of four reef fish species in Puerto Rico. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 50: 938–968.
- Figueroa, M. 1991. Aspectos reproductivos del chillo (*Lutjanus vivanus*) en el oeste de PR y sus implicaciones en el manejo pesquero. 20 p.

- Franks, J.S., N.J. Brown-Peterson, M.S. Griggs, N.M. Garber. 2000. Potential of the first dorsal fin spine for estimating the age of wahoo, *Acanthocybium solandri*, from the northern Gulf of Mexico, with comments on specimens from Bimini, Bahamas. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 51: 682.
- Gobert, B., A. Gillou, P. Murray, P. Berthou, M.D. Oqueli Turcios, E. Lopez, P.L. Jérôme Huet, N. Diaz, P. Gervain. 2005. Biology of queen snapper (*Etelis oculatus*: Lutjanidae) in the Caribbean. *Fish. Bull.* 103: 417–425.
- Jones, D.D., J.M. Rivera Hernández, V.R. Shervette. 2021. Princess parrotfish *Scarus taeniopterus* age, growth, maturity, and transition. *Environ Biol Fish*, 104: 581–595.
- Kadison, E., R. Nemeth, N. Brown-Peterson, J. Blondeau, T. Smith, J. Calnan. 2011. Yellowfin grouper (*Mycteroperca venenosa*): reproductive biology, behavior and conservation of a large Caribbean grouper. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 63: 157–160.
- Luckhurst, B.E., J.M. Dean, M. Reichert. 2000. Age, growth and reproduction of the lane snapper *Lutjanus synagris* (Pisces: Lutjanidae) at Bermuda. *Mar Ecol Prog Ser*, 203: 255–261.
- Luckhurst, B.E., J.A. Barnes, Y. Sadovy. 1992. Record of an unusually large red hind, *Epinephelus guttatus* (Pisces: Serranidae) from Bermuda with comments on its age. *Bulletin of Marine Science*, 51(2): 267–270.
- Maki Jenkins, K.L. & R.S. McBride. 2009. Reproductive biology of wahoo, *Acanthocybium solandri*, from the Atlantic coast of Florida and the Bahamas. *Marine and Freshwater Research*, 60(9): 893–897.
- Manickchand-Dass, S. 1987. Reproduction, age and growth of the lane snapper, *Lutjanus synagris* (Linnaeus), in Trinidad, West Indies. *Bulletin of Marine Science*, 40(1): 22–28.
- Manooch, C.S., III & C.L. Drennon. 1987. Age and growth of yellowtail snapper and queen triggerfish collected from the U.S. Virgin Islands and Puerto Rico. *Fisheries Research*, 6(1): 53–68.
- Mateo, I. & W.J. Tobias. 1999. Preliminary estimation of growth, mortality and yield per recruit for the spiny lobster, *Panulirus argus*, in St. Croix, USVI. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 53: 59–75.
- Molloy, P.P., M.J. Paddock, J.D. Reynolds, M.J.G. Gage, I.M. Côté. 2011. Relative size-at-sex-change in parrotfishes across the Caribbean: is there variance in a supposed life-history invariant? *Evol Ecol*. 19 p.
- Mueller, K.W. 1995. Size structure of mutton snapper, *Lutjanus analis*, associated with unexploited artificial patch reefs in the central Bahamas. *Fishery Bulletin*, 93: 573–576.
- Munro, J.L. & D.M.B. Williams. 1985. Assessment and management of coral reef fisheries: biological, environmental, and socio-economic aspects. *Proceedings of the Fifth International Coral Reef Congress*, Tahiti, 4: 545–581.
- Nemeth, R.S., E. Kadison, N.J. Brown Peterson, J. Blondeau. 2020. Reproductive biology and behavior associated with a spawning aggregation of the yellowfin grouper *Mycteroperca venenosa* in the US Virgin Islands. *Bulletin of Marine Science*, 69(1): 31–56.
- Nemeth, R.S. 2005. Population characteristics of a recovering US Virgin Islands red hind spawning aggregation following protection. *Mar Ecol Prog Ser*, 286: 81–97.
- Olsen, D.A., J. Nowlis, D. Bryan. 2017. A Study of the Virgin Islands Spiny Lobster Fishery: Growth, Population Size, and Mortality. SEDAR57-RD-10. 28 p.
- Olsen, D.A. 2014. Population Size, Growth, Mortality and Movement Patterns of Yellowtail Snapper (*Ocyurus chrysurus*) in the U.S. Virgin Islands Determined Through a Multi-institutional Collaboration. SEDAR84-RD-03. 55 p.
- Oxenford, H.A., P.A. Murray, B.E. Luckhurst. 2003. The Biology of Wahoo (*Acanthocybium solandri*) in the Western Central Atlantic. *Gulf and Caribbean Research*, 15: 33–49.
- Pagan, F.E. 2002. Caribbean-SEAMAP Program Reef Fish data Analysis Survey. Final Report.
- Pérez, R.N. & A.M. Roman. 2005. Tournament fishing effort estimates and reproductive dynamics of the dolphinfish, *Coryphaena hippurus*, L. in Puerto Rico. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 47: 705–743.
- Pérez, R.N. & Y. Sadovy. 1991. Preliminary data on landings records and reproductive biology of *Coryphaena hippurus* L., in Puerto Rico. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 44: 651–670.
- Pérez R.N., A.M. Román, G.A. Rivera. 1992. Investigation on the reproductive dynamics and preliminary evaluation of landings data of dolphinfish *Coryphaena hippurus*, L. Final Report for Dingell-Johnson

- Project F26-1. Puerto Rico Dept. of Natural Resources Fishery Research Laboratory, Mayaguez, PR. 95 p.
- Pérez-Chacón, J.E.B. & A. Aguilar-Perera. 2015. Length–weight and length–length relationships of the Hogfish, *Lachnolaimus maximus*, off the northern coast of the Yucatan Peninsula, Mexico. *Gulf and Caribbean Research*, 26(1): 3 p.
- Posada J.M. & R. Glazer. 2007. Biology and fisheries of the spiny lobster (*Panulirus argus*) and queen conch (*Strombus gigas*) based on the Proceedings of the Gulf and Caribbean Fisheries Institute. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 60: 54–61.
- Randall, J.E. 1962. Tagging reef fishes in the Virgin Islands. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 14: 201–241.
- Rivera, G.A. & R.S. Appeldoorn. 2000. Age and growth of dolphinfish, *Coryphaena hippurus*, off Puerto Rico. *Fishery Bulletin*, 98: 345–352.
- Rivera Betancourt, G.A. 1994. Age and growth of dolphinfish, *Coryphaena hippurus* L., in Puerto Rico as determined by otolith analysis. University of Puerto Rico, Mayaguëz Campus.
- Rivera-Hernández, J.M. & V.R. Shervette. 2024. Addressing life history information gaps for Caribbean parrotfishes: queen parrotfish *Scarus vetula* and stoplight parrotfish *Sparisoma viride*. *Environmental Biology of Fishes*, 108: 179–198.
- Rivera-Hernández, J.M. & V.R. Shervette. 2024. Puzzling parrotfishes: Radiocarbon age validation and updated longevity estimates for western Atlantic species in support of sustainable fisheries management. *PLoS ONE*, 19(5): e0302854.
- Rivera-Hernández, J.M. & V.R. Shervette. 2024. Queen Triggerfish *Balistes vetula* Age-Based Population Demographics and Reproductive Biology for Waters of the North Caribbean. *Fishes*, 9(5): 162. <https://doi.org/10.3390/fishes9050162>
- Rivera-Hernández, J.M. & V.R. Shervette. 2022. Queen Triggerfish *Balistes vetula* Age-Based Population Demographics and Reproductive Biology for Waters of the North Caribbean. *Fishes*, 9(5): 162.
- Rivera-Hernández, J.M., N. Peña Alvarado, K.C. Vélez, R. Nemeth, R. Appeldoorn, V. Shervette. 2018. Queen Triggerfish Reproductive Biology in U.S. Caribbean Waters. *Transactions of the American Fisheries Society*, 148(1): 134–147.
- Román, A. 1991. Estudio sobre la dinámica reproductiva de la cachicata blanca: *Haemulon plumieri*. MS Thesis Dept. Marine.
- Rosario, A. J. Rojas, E. Piñeiro, M. Figuerola, N. Peña, W. Torres. 2006. Reproductive Cycle and Maturation Size of Silk Snapper (*Lutjanus vivanus*). Final Report to Caribbean Fishery Management Council. 21 p.
- Rosemond, R.C., R.S. Nemeth, S.A. Heppell. 2022. Demographic Recovery of a Reef Fish Population Over 30 Years of Spawning Aggregation Site Protection. *Frontiers in Marine Science*, 9: 931409.
- Sadovy, Y., M. Figuerola, A. Román. 1989. The age and growth of the red hind *Epinephelus guttatus* and the white grunt *Haemulon plumieri* in Puerto Rico and the U.S. Virgin Islands. Final Report to the Caribbean Fishery Management Council. 69 p.
- Sadovy, Y., M. Figuerola, A. Román. 1992. Age and growth of red hind *Epinephelus guttatus* in Puerto Rico and St. Thomas. *Fishery Bulletin*, 90: 516–528.
- Sadovy, Y., A. Rosario, A. Román. 1994. Reproduction in an aggregating grouper, the red hind, *Epinephelus guttatus*. *Environmental Biology of Fishes*, 41(1): 269–286.
- Shapiro, D.Y., Y. Sadovy, M.A. McGehee. 1993. Size, Composition, and Spatial Structure of the Annual Spawning Aggregation of the Red Hind, *Epinephelus guttatus* (Pisces: Serranidae). *Copeia*, 1993(2): 399–406.
- Shapiro, D.Y., Y. Sadovy, A.M. McGehee. 1993. Periodicity of Sex Change and Reproduction in the Red Hind, *Epinephelus guttatus*, a Protogynous Grouper. *Bulletin of Marine Science*, 53(3): 1151–1162.
- Shervette, V.R. & J.M. Rivera-Hernández. 2022. Queen triggerfish *Balistes vetula*: Validation of otolith-based age, growth, and longevity estimates via application of bomb radiocarbon. *PLoS ONE*, 17(1): e0262281.
- Shervette, V.R. & J.M. Rivera-Hernández. 2024. U.S. Caribbean Yellowtail Snapper Population Demographics, Growth, and Reproductive Biology: Addressing Critical Life History Gaps. SEDAR84-DW-08. 31 p.
- Stevens, M.H., S.G. Smith, J.S. Ault. 2019. Life history demographic parameter synthesis for exploited Florida and Caribbean coral reef fishes. *Fish and Fisheries*, 20: 1196–1217.

- Stevenson, D.K. 1978. Management of tropical pot fishery for maximum sustainable yield. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 30: 95–115.
- Sylvester, J.R. & A.E. Dammann. 1973. Contribution to the Biology of the Silk Snapper *Lutjanus vivanus* from the Virgin Islands. *Transactions of the American Fisheries Society*, 102(4): 843–845.
- Tewfik, A. & E.A. Babcock. 2017. Examining Caribbean spiny lobster size at maturity using external reproductive-related structures in Belize. *Proceedings of the Gulf and Caribbean Fisheries Institute*, 70: 220–221.
- Trumble, R.J., D. Olsen, N. Cummings. 2006. A pilot program to assess methods of collecting bycatch, discard, and biological data in the commercial fisheries of St. Thomas, U.S. Caribbean. SEDAR30-RD-01. 81 p.
- van Rooij, J.M. & J.J. Videler. 1997. Mortality estimates from repeated visual censuses of a parrotfish (*Sparisoma viride*) population: demographic implications. *Marine Biology*, 128: 385–396.
- Whiteman, E.A., C.A. Jennings, R.S. Nemeth. 2005. Sex structure and potential female fecundity in a *Epinephelus guttatus* spawning aggregation: applying ultrasonic imaging. *Journal of Fish Biology*, 66: 983–995.